

Omer Pinto

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Summary & Current Focus

- Senior Software & **AI Engineering leader** with **15+ years** building **backend systems, database internals, & distributed platforms**.
- Strong foundation in **algorithms, performance tuning & systems architecture across on-prem & cloud platforms**.
- Currently **AI Enablement Lead** at BigArch, driving **organization-wide AI adoption** across developer workflows, **coding agents**, context engineering, evaluation, and **AI-enabled SDLC transformation**, while remaining **hands-on** in prototyping and building internal AI platforms.
- Seeking **AI Engineering roles** focused on intelligent products and **LLM-powered applications**, or **AI Enablement leading roles**.

Experience

BigArch

AI ENABLEMENT LEAD

2026 - Present

- Initiated the organization's **AI adoption program** and established its **AI Champions model**, defining strategy, methodology, goals, recurring processes, and collaboration with engineering teams.
- Created and delivered hands-on **Claude Code workshops, technical training**, and coaching for **~100 engineers**.
- Improve **DevEx** by designing reusable **Claude Code playbooks and skills** to support development pipelines, while evaluating and integrating external tools that improve the Claude Code developer experience.
- Conceived and built an **LLM-powered adoption intelligence platform** that transforms unstructured team updates into structured organizational knowledge while preserving historical progress.
- Explore **adoption measurement** approaches & internal tooling to track progress, surface blockers, and assess **AI-in-SDLC** initiatives.
- Evaluate **context-engineering** techniques, models, and **agentic approaches** across diverse use cases, adapting them for effective operation in an **air-gapped environment**.

SENIOR SW ENGINEER

2024 - 2026

- Built **backend services** for a large-scale **graph orchestration platform** (HPC, **Python, MongoDB, RabbitMQ**, microservices); deployed containerized services in private-cloud via **Kubernetes & Docker**.
- Achieved **60% performance gains** through system-wide optimization; led architectural refactors executed with **zero downtime** and full service continuity.
- Independently modernized a legacy React application with **Claude Code: 23K+ LOC rewritten, 238 files** migrated to strict TypeScript, full framework upgrade, and **zero regressions**.

Microsoft

SENIOR SW ENGINEER

2023 - 2024

- Engineer in the **vulnerability management** group, developing components in the endpoint security pipeline.
- Built **C# microservices** and data pipelines using Azure services (Blobs, Queues, CosmosDB, KustoDB).

Yahoo

PRINCIPAL SW ENGINEER

2022 - 2023

- Contributor to the **native ads** platform, developing **Java** services and **Hadoop/Spark** ETL pipelines.
- Partnered with research team to implement **ML/DL-driven ad-scoring algorithms** for multi-variable ranking system.
- Managed codebase migration and team handoff following **project cancellation**.

qSpark

SENIOR SW ENGINEER

2021 - 2022

- Delivered **ultra-low-latency trading** infrastructure. Implemented **exchange protocols** (execution & market data) using **modern C++ with template meta-programming**.
- Designed **scalable** subsystem enabling 3rd-party **trading algorithms** integration.
- Handled production issues with **global exchanges** and conducted market-structure research (SEC/FINRA rules).

Toga Networks

HANDS-ON ARCHITECT

2019 - 2021

- Worked across multiple projects in the **cloud and IoT R&D** group, driving architecture and implementation.
- Performed **market and technology research** and delivered high-level designs for innovative systems.
- Led research & design of semantic analytics engine for IoT data, including reasoning capabilities across data model and query layers.
- Developed a log-analysis algorithm using **exponential backoff**, improving system performance.

SiSense

SENIOR ENGINEER, TECH LEAD & ARCHITECT, CTO OFFICE

2014 - 2018

- **Hand-picked by CTO** for his own private team researching next-gen analytics & developing features on open-source DB project.
- Architected and built cross-platform **ETL query engine from scratch** as sole developer; recruited and led team as tech lead.
- Engine optimized for **low RAM footprint**, 100% multicore **CPU utilization** (any scale-up) and **distributed execution** (scale-out), implemented using state-of-the-art **C++14/17 template meta-programming**.
- Implemented **core DB features** and system optimizations using multicore algorithms and **cache-aware data structures**.
- Designed thread management solution **improving** query engine performance **by order of magnitude**.
- Optimized query operations (join, group) on multi-billion record datasets.
- Implemented statistical **algorithms from academic papers**.

Early years - Allot Communications, ZRRO Technologies

FIRMWARE / EMBEDDED ENGINEER

2011-2014

- Design & implementation of **real-time algorithms & data structures** in a resource-restricted, **performance-critical** environment.
- Network protocols, coding of algorithms & systems in **C & C++** using extreme **memory/cpu optimizations** on **Linux** os.

Technical Skills

- **Modern C++ (11/14/17/20)**: Expert-level proficiency including STL, Boost, Intel TBB. Advanced template metaprogramming.
- **Python & Backend Development**: Production experience with microservices architecture, distributed systems (MongoDB, RabbitMQ, Kafka). Async programming, performance optimization. Docker, Kubernetes for containerized deployments.
- **AI & LLM Systems**: Design and build **LLM applications** using LangGraph, OpenAI Agents SDK, Claude Agent SDK, **MCP**, structured outputs, tool calling, guardrails, evaluation, and human-in-the-loop workflows.
- **AI-Native Product Development**: Identify high-value engineering gaps, rapidly prototype solutions with **Claude Code**, and evolve them into sophisticated internal platforms using reusable **skills, sub-agents, parallel orchestration**, and deterministic execution workflows.
- **AI Engineering Enablement**: **AI-in-SDLC** strategy, Developer Experience, organization-wide training, adoption measurement, continuous evaluation of open-source models in **air-gapped environments**, and delivering internal AI platforms as adoption solutions.
- **Database Systems & Internals**: Deep expertise in DB internals (query parsers, execution engines, optimizers, caching layers). Designed and built ETL engine from scratch. Working knowledge of PostgreSQL, MongoDB, ClickHouse. Query language DSL design.
- **Performance Engineering**: Low-latency algorithms, memory/CPU profiling, cache-aware data structures, multi-threading.
- **Additional Languages**: Java (Spring, Maven), C# (Azure microservices).
- **Cloud Platforms**: Azure services (Blobs, Queues, CosmosDB, KustoDB), AWS (EC2, S3, Lambda).
- **Big data analytics**: Implement Hadoop/Spark data pipelines for cross-organization ETL workflows.
- **Distributed graph systems**: Built solutions for on-prem and HPC environments, coordinating millions of tasks.

Education

The Open University of Israel

Ra'anana, Israel

B.SC IN MATHEMATICS AND COMPUTER SCIENCE

2007 - 2010

- Graduated **summa cum laude** with GPA of **97/100**.
- Awarded place on University President's Excellence List (annual GPA ≥ 95) for four consecutive years.
- Achievement recognized in university educational portal [article](#).

Independent Work & Technical Writing

AI Projects, Udemy AI Learning & Technical Writing

- **Technical C++ writing**: Published a series of articles on database implementation & C++ techniques on LinkedIn, republished by [EuropeClouds](#) (2020-2021). Created [C++ TMP Tutorial](#) repo referenced in articles.
- **LangGraph Infrastructure**: Created Python abstraction layer providing reusable components (nodes, edges, tools, structured outputs) for agent development. [GitHub](#).
- **Agentic Workflows**: Implement multiple projects (stock market predictor, real-time soccer betting system) utilizing my infra project & external MCP integrations. [Stocks, Betting](#).
- **The Complete Agent & MCP Course** (Udemy): completed a comprehensive training in AI agents platforms.
- **From Vibe Coder to Agentic Engineer** (Udemy, Ed Donner): deep dive on **Claude Code**, multi-agent workflows, and context engineering patterns.